Lecture 1: Segmentation

From U-Net to nnU-Net — one label per pixel

Mehmet Kurt, Tianyi Ren
University of Washington

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Segmentation is a dense classification task

- Classification: **one label per image**.
- Segmentation: **one label per pixel**.

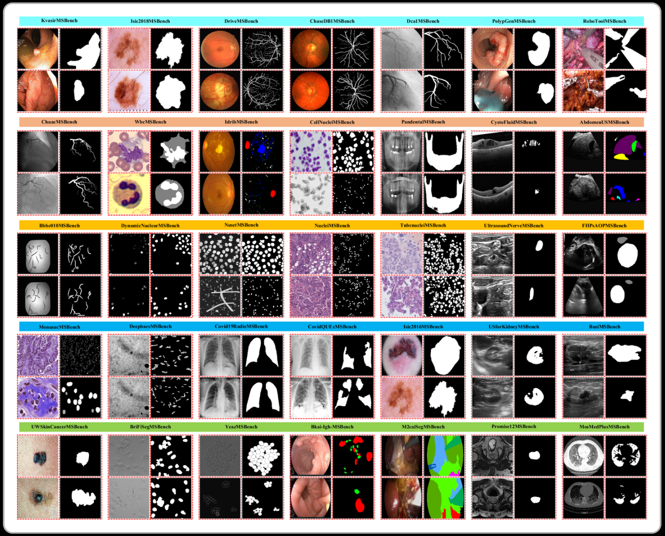
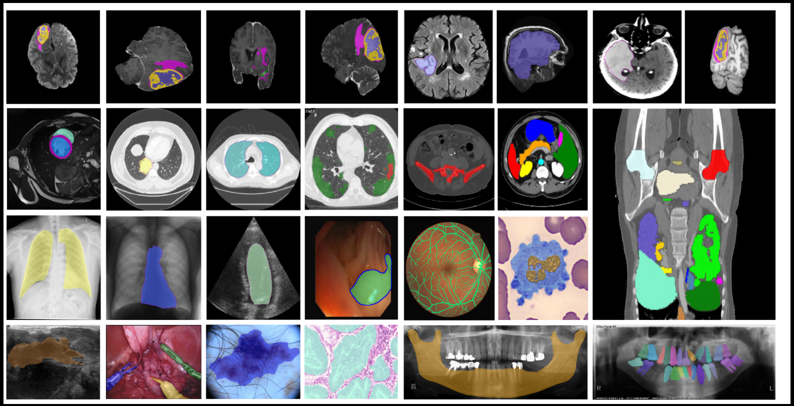


Fig. 1, Kuş & Aydin, *MedSegBench*, Scientific Data **11**, 1283 (2024), CC BY-NC-ND 4.0.

Why clinicians want masks, not labels

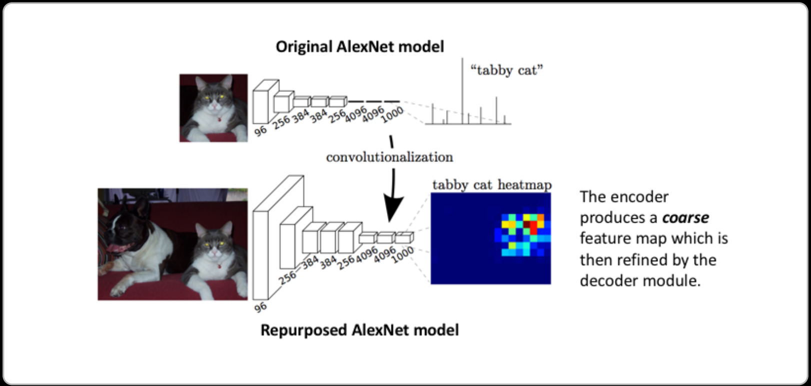


Medical image segmentation is used for **diagnosis**, **treatment planning**, and **patient care** — because a label cannot be measured, and a mask can.

Figure 1 and clinical applications: Bougourzi & Hadid, Med. Image Anal. 2025 (arXiv:2505.09274).

The U-Net

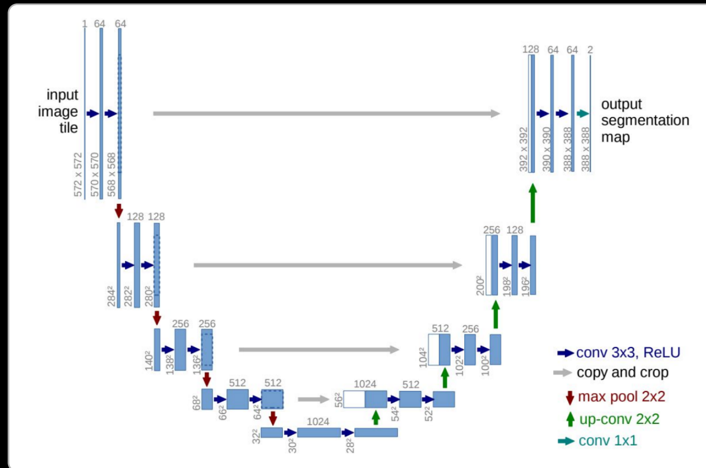
THE RESOLUTION PROBLEM



Segmentation needs a **full-resolution** output.

Figure: J. Jordan, *An overview of semantic segmentation* (jeremyjordan.me), modified from Long, Shelhamer & Darrell, *Fully Convolutional Networks for Semantic Segmentation*, CVPR 2015.

The U-Net



Contracting path

Two 3×3 **unpadded convolutions** + ReLU, then 2×2 **max pooling**, stride 2. Each step **doubles the feature channels**.

Expansive path

Upsampling, then a 2×2 **up-convolution** that **halves the feature channels**, and two 3×3 convolutions + ReLU.

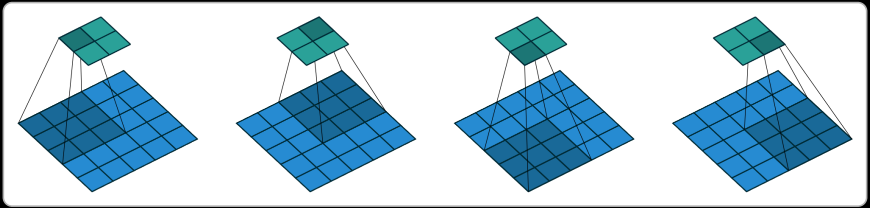
Copy and crop

Concatenation with the **correspondingly cropped** feature map from the contracting path — every unpadded convolution loses border pixels.

Terminology after Ronneberger, Fischer & Brox, *U-Net: Convolutional Networks for Biomedical Image Segmentation*, MICCAI 2015 (arXiv:1505.04597), §2.

Convolution (2D)

A 3×3 kernel over a 5×5 input, stride 2 — output 2×2.



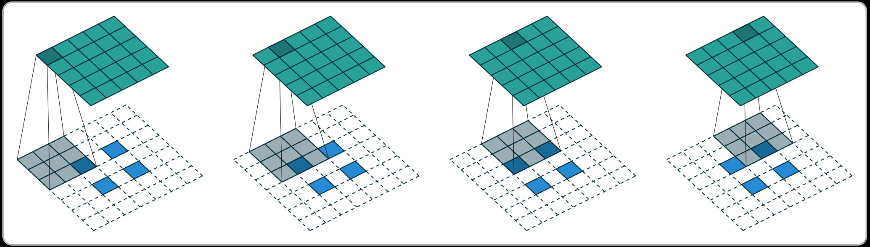
$$y[r,c] = \sum_{m=0}^{k-1}\sum_{n=0}^{k-1} x\left[\,s\,r+m,\ s\,c+n\,\right]\, w[m,n]$$

output size $\quad o = \left\lfloor \dfrac{i+2p-k}{s} \right\rfloor + 1 = \left\lfloor \dfrac{5+0-3}{2} \right\rfloor + 1 = 2$

Figure 2.5 from Dumoulin & Visin, *A guide to convolution arithmetic for deep learning* (arXiv:1603.07285). Input blue, output green.

Transposed convolution

The transpose of that convolution: 2×2 in, 5×5 out.



$$y\big[\,s\,r + m,\ s\,c + n\,\big] \ \mathrel{+}= \ x[r, c]\ w[m, n]$$

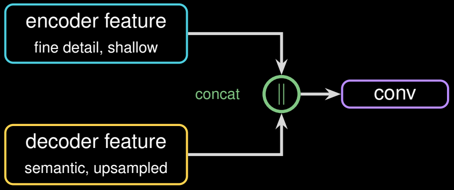
output size $\quad o = s\,(i-1) + k - 2p = 2\,(2-1) + 3 - 0 = 5$

$y = C^{\top} x \quad$ — hence *transposed*.

Figure 4.5 from Dumoulin & Visin (arXiv:1603.07285) — the transpose of Figure 2.5 on the previous slide.

Skip connections: where the boundaries come from

- Upsample the coarse decoder feature, then **concatenate** the matching encoder feature map.
- The encoder map still holds the *fine spatial detail* that pooling destroyed.
- Result: semantics from the deep path, **precise boundaries** from the shallow one.
- Bonus: a short gradient path to the early layers, exactly as in ResNet.



Concatenate, not add — the decoder learns how to weigh the two sources.

How do we know it worked?

IoU (Jaccard index)

Intersection over union

$$\text{IoU} = \frac{TP}{TP + FP + FN}$$

Pixels in *both* masks, over pixels in *either*. Range $[0, 1]$, higher is better.

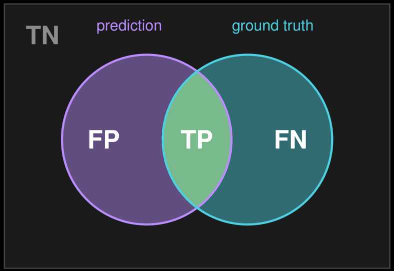
Work one through

Prediction 100 px, ground truth 120 px, overlap 80 px.

TP=80, FP=20, FN=40.

$\text{IoU} = 80/140 = \mathbf{0.57}$ Dice $= 160/220 = \mathbf{0.73}$

Compute **per image**, then average — not pooled over the test set, which lets big easy cases hide small hard ones.



TN never appears — which is exactly why IoU survives class imbalance and accuracy does not.

Dice coefficient

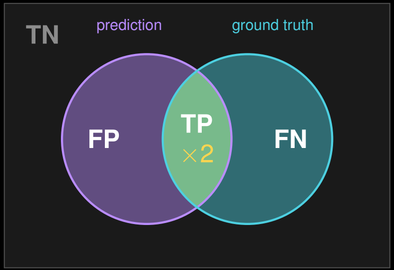
The metric medical imaging reports

$$\mathrm{DSC} = \frac{2\,TP}{2\,TP + FP + FN}$$

Twice the intersection, over the sum of the two mask sizes — and again, no TN.

Also known as

The **F1 score** computed over pixels: precision $= TP/(TP+FP)$ and recall $= TP/(TP+FN)$, harmonic mean. Also called DSC, the Dice similarity coefficient.



TP is counted **twice** in the numerator — that is the whole difference from IoU.

Dice vs. IoU: never argue about which is better

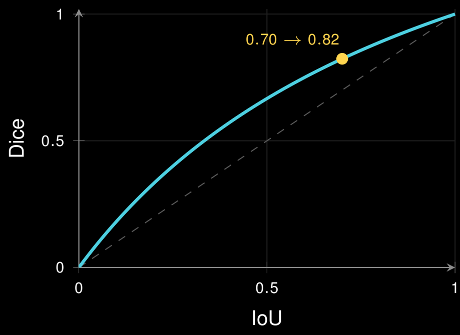
They are the same ranking

$$\text{Dice} = \frac{2\,\text{IoU}}{1 + \text{IoU}} \qquad \text{IoU} = \frac{\text{Dice}}{2 - \text{Dice}}$$

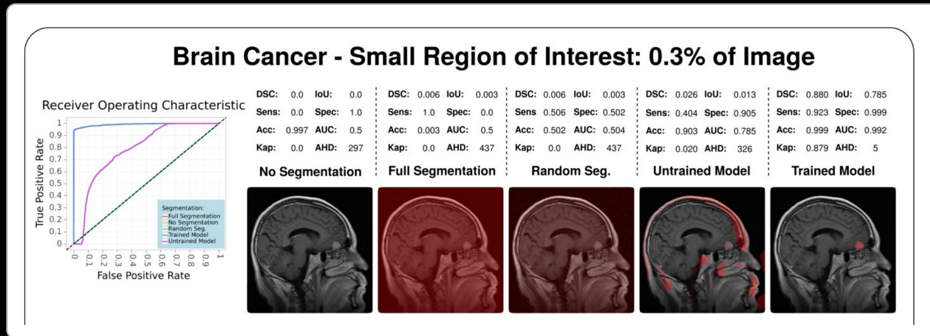
A strictly increasing map. If model A beats B on Dice, it beats B on IoU.

But the numbers differ

Dice is **never smaller** than IoU, and strictly larger whenever the match is imperfect. IoU 0.70 is Dice 0.82. Check which one a paper quotes before you compare against it.



Accuracy is highly discouraged



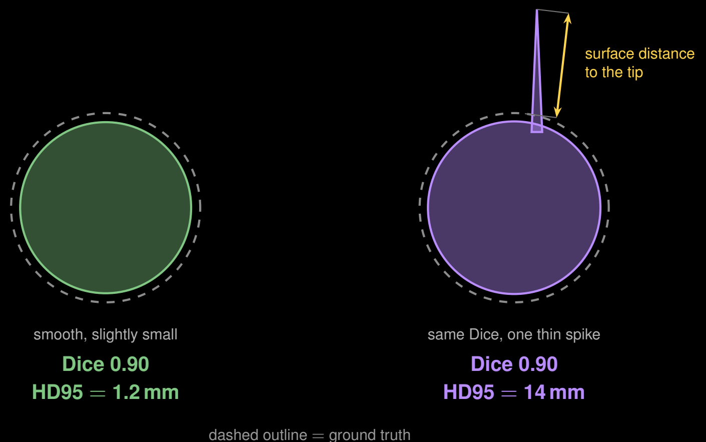
From finding **nothing** to finding the tumour, accuracy moves **0.997** to **0.999**. Dice moves **0.000** to **0.880**.

Dice and IoU are both appropriate — accuracy is not.

Müller, Soto-Rey & Kramer, arXiv:2202.05273 (2022), Fig. 1 upper panel; wording paraphrased.

Overlap is not enough

Dice and IoU measure **overlap, not boundary accuracy**. Two predictions, **identical Dice**:



The spike is **2.7% of the area**, so Dice barely moves — but **31% of the boundary**.

Both panels are constructed to score Dice 0.90 against the same ground truth. Point after VeraDP, *Dice Coefficient vs. IoU in Medical Image Segmentation*, citing Kocak et al. 2025.

Hausdorff distance and HD95

Distance-based measures score the **spatial distance between predicted and ground-truth boundaries**.

$$\mathrm{HD} = \max\left(\sup_{p \in P} \inf_{g \in G} d(p, g), \ \sup_{g \in G} \inf_{p \in P} d(g, p) \right)$$

the maximum distance from a boundary point of one set to the closest point of the other — the **worst-case** boundary error

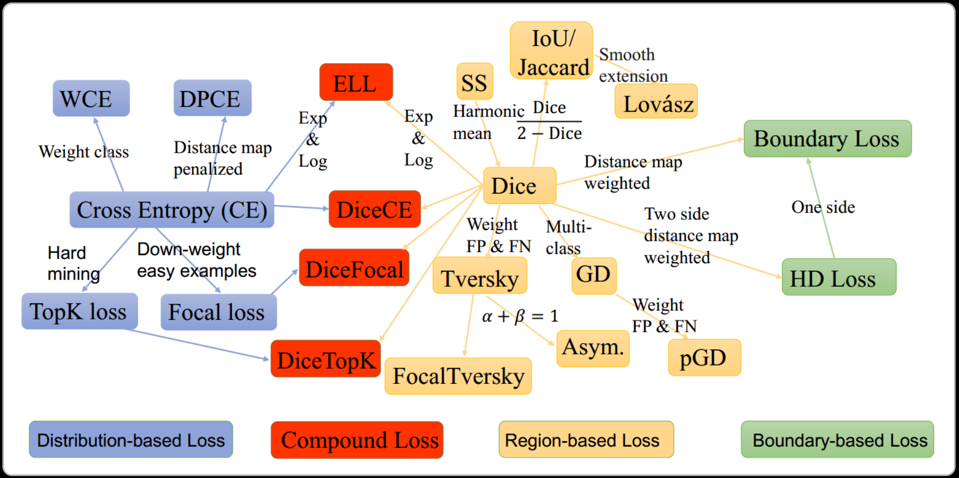
HD95 — the maximum is susceptible to outliers, so report the **95th percentile** instead.

It is measured in millimetres: resample to a common voxel spacing before comparing.

P, *G* are the predicted and ground-truth boundary point sets. HD: Hausdorff (1914); HD95: Taha & Hanbury (2015). HD95 is the standard boundary metric in BraTS and the Decathlon.

Training it

THE LOSS: CLASSIFY EVERY PIXEL



Distribution-based, region-based, boundary-based — and the compound losses that mix them.

Loss taxonomy: Ma, Chen, Ng, Huang, Li, Li, Yang & Martel, *Loss odyssey in medical image segmentation*, Medical Image Analysis **71** (2021) 102035; figure from the authors' repository, github.com/JunMa11/SegLoss.

Cross-entropy vs. Dice loss

Pixelwise cross-entropy

Treats every pixel as an independent classification.

Good: smooth, well-behaved gradients everywhere.

Bad: a tiny lesion is drowned by background, and the cheapest way to lower the loss is to predict background everywhere.

Dice loss

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}$$

Optimises the metric directly, using soft probabilities p_i so it stays differentiable.

Good: scale-invariant, so a small structure still matters.

Bad: noisy gradients when the structure is absent or tiny.

The ϵ is not cosmetic — it is what keeps an empty ground-truth slice from dividing by zero.

In practice: use both

The default

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}$$

CE supplies dense, stable gradients early in training; Dice stops the model from collapsing to background. This is what nnU-Net uses, and it is a genuinely hard baseline to beat.

If it still collapses

Class weights in the CE term, or **focal loss** to down-weight the easy background pixels.

If boundaries leak

Add a boundary-aware term. But first check it is not simply a **resolution** problem — often the mask is fine and the output stride is too coarse.

Track **validation Dice**, not validation loss. They are not the same curve, and only one of them is the thing you care about.

Training loop, concretely

The standard training loop; only the model, loss and metric change:

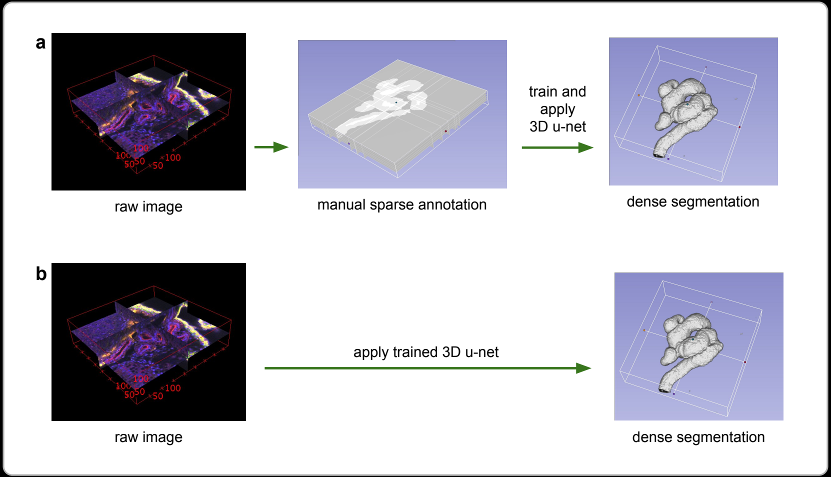
```
for x, mask in loader:
  logits = unet(x)
  loss = bce(logits, mask) \
     + dice_loss(logits, mask)
  optimizer.zero_grad()
  loss.backward()
  optimizer.step()
# predict:  sigmoid(logits) > 0.5
```

- Output is a **per-pixel logit map**.
- Augment **image and mask together** — a flip that misses the mask silently corrupts the label.
- Heavy augmentation: flips, rotations, scaling, elastic deformation. Masks are scarce.
- Validate on **Dice**, and keep the best-Dice checkpoint.

Going 3D

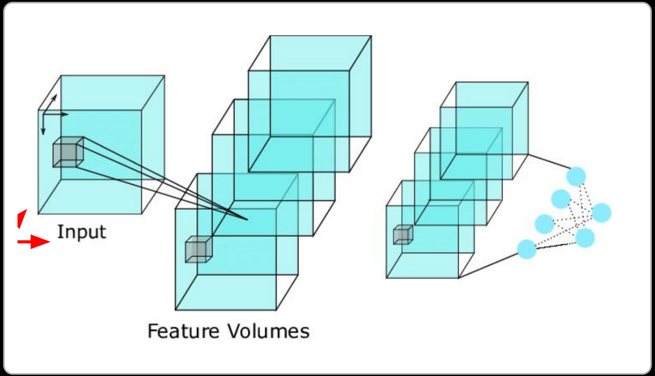
MEDICAL IMAGES ARE VOLUMES, NOT PICTURES

- Volumetric data is abundant in biomedical data analysis. Thus, annotation of large volumes in a slice-by-slice manner is **very tedious**.
- Annotation of such data with segmentation labels causes difficulties, since only 2D slices can be shown on a computer screen.



Çiçek et al., *3D U-Net*, MICCAI 2016, Fig. 1 — (a) semi-automated, (b) fully-automated.

From 2D to 3D convolution



2D convolution

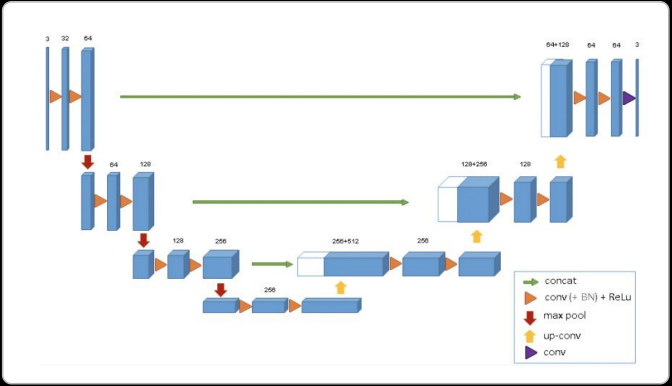
A $k \times k$ filter slides over height and width, and spans the *full* channel depth. Two directions of movement.

3D convolution

A $k \times k \times k$ filter slides along x, y **and** z. Three directions — depth is now a spatial axis, not a channel axis.

3D U-NET

While the u-net is an entirely 2D architecture, the network proposed in this paper takes 3D volumes as input and processes them with corresponding 3D operations, in particular, 3D convolutions, 3D max pooling, and 3D up-convolutional layers.



Analysis path

Four resolution steps. Each layer contains two $3\times3\times3$ convolutions each followed by a ReLU, and then a $2\times2\times2$ max pooling with strides of two in each dimension.

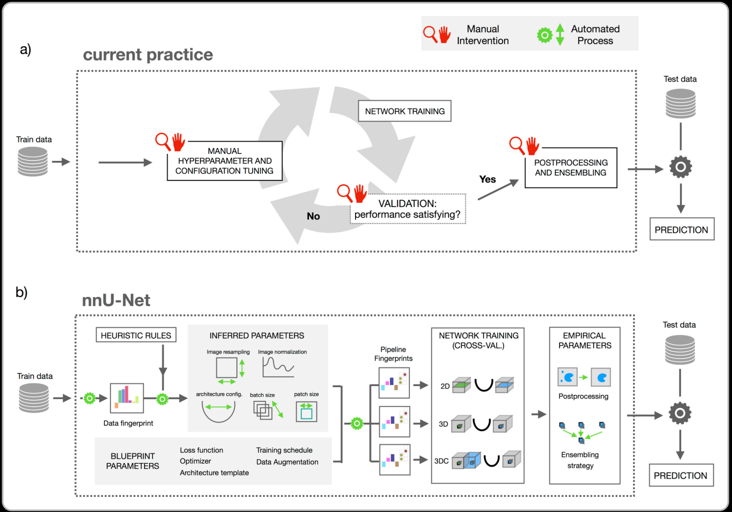
Synthesis path

Each layer is an upconvolution of $2\times2\times2$ by strides of two, followed by two $3\times3\times3$ convolutions each followed by a ReLU. Shortcut connections from layers of equal resolution in the analysis path provide the **essential high-resolution features**.

19,069,955 parameters in total. — Çiçek et al., *3D U-Net*, MICCAI 2016.

nnU-Net: stop designing architectures

NNU-NET: CONFIGURE THE PIPELINE, NOT THE NETWORK



(a) current practice

A manual trial-and-error loop: tune, train, look at validation, tune again. The red hands are the human.

(b) nnU-Net

Fingerprint the data, run the rules, train 2D / 3D / cascade, pick the ensemble. **No hands.** It uses a plain U-Net — hence *no new net*.

Figure 2 from Isensee, Jaeger, Kohl, Petersen & Maier-Hein, *Automated Design of Deep Learning Methods for Biomedical Image Segmentation* (arXiv:1904.08128) — the open preprint of *nnU-Net: a self-configuring method. . .*, Nature Methods **18**:203–211, 2021.

What nnU-Net actually decides

Fixed	same for every dataset	plain U-Net, loss = CE + Dice, SGD with Nesterov momentum, deep supervision (a loss at every decoder level), the augmentation set
Rule-based	computed from the fingerprint	patch size, batch size, network depth and downsampling, target spacing and resampling, intensity normalisation
Empirical	chosen by cross-validation	2D vs 3D vs 3D-cascade, which models to ensemble, whether to apply connected-component postprocessing

TRAINED FROM SCRATCH ON CHALLENGE DATA ALONE

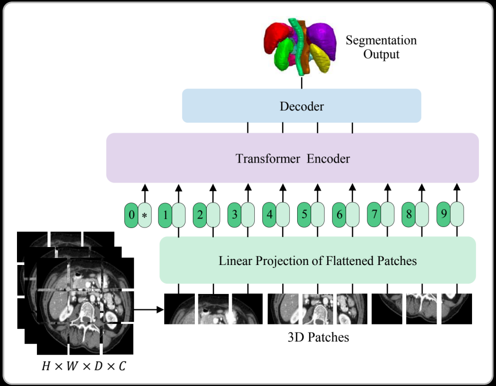
33/53 new state of the art **23** datasets **11** challenges

Elsewhere, on par with or close to the top entry — organs, tumours and cells, in MRI, CT, electron and fluorescence microscopy.

Isensee, Jaeger, Kohl, Petersen & Maier-Hein, Nature Methods **18**:203–211, 2021.

Other transformers in medical image segmentation

UNETR: A TRANSFORMER ENCODER

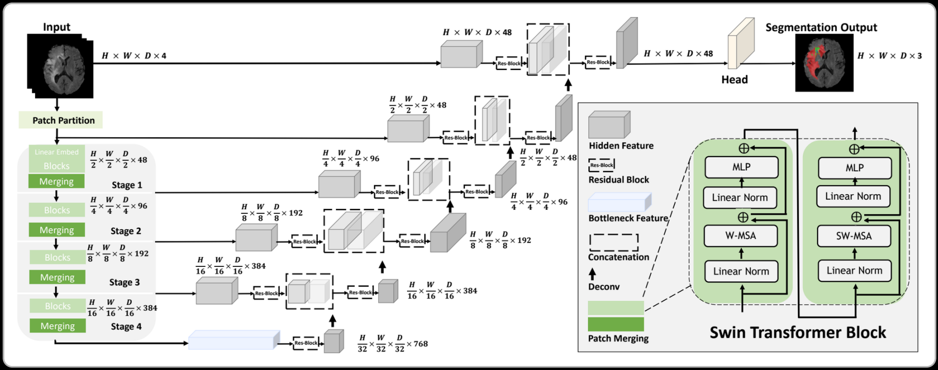


- Convolution is **local.**
- The volume becomes **3D patches.**
- A **transformer** encodes them.
- A CNN decoder rebuilds the mask.

Only the encoder changed.

Figure 1, Hatamizadeh et al., *UNETR: Transformers for 3D Medical Image Segmentation*, WACV 2022 (arXiv:2103.10504).

SWIN UNETR: ATTENTION IN SHIFTED WINDOWS

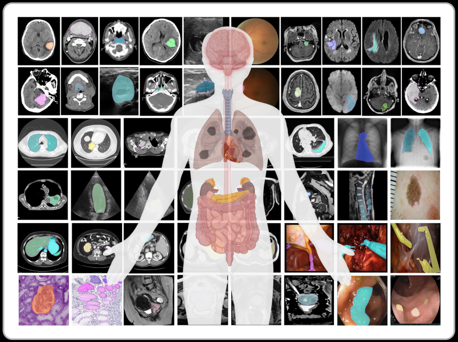


Attention inside **shifted windows**. Four stages, five resolutions, same CNN decoder.

Built for BraTS, where tumours vary in size.

Figure 1, Hatamizadeh et al., *Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images*, BrainLes/MICCAI 2021 (arXiv:2201.01266).

Or do not train at all: MedSAM



- One model, any modality.
- Prompt it with a **bounding box**.
- **No training.**

1.57M image–mask pairs
10 modalities
30+ cancer types

Expert annotation time — 82% less.

Figure 1, Ma et al., *Segment anything in medical images*, Nature Communications **15**, 654 (2024), CC BY 4.0.